# Effectiveness of Artificial Intelligence-Based Adaptive Honeypots in Cyber Threat Detection: A Systematic Literature Review and Meta-Analysis

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***Abstract*—This study conducted a systematic literature review and meta-analysis on using honeypot systems enhanced by artificial intelligence to improve the effectiveness of cyber threat detection in organizational environments. The review followed the PRISMA protocol and assessed 454 articles from databases, including IEEE, ACM, Emerald, and Web of Science. After a multi-stage screening process, 62 articles met the inclusion criteria and were further analyzed. The synthesis indicated that integrating artificial intelligence into honeypot systems improved detection accuracy, expanded the system's ability to recognize varied attack patterns, and optimized resource efficiency. A meta-analysis of 36 studies revealed consistent, significant improvements in detection performance. Quantitatively, the analysis yielded a mean effect size of 0.905, indicating a substantial improvement in detection effectiveness resulting from AI integration. These findings confirm that adopting AI-based honeypot technologies is essential for addressing increasingly complex cyberattacks and for providing a foundation for future research into the development of standardized evaluation frameworks**.

***Keywords—Honeypot, Artificial Intelligence, Intrusion Detection System, Systematic Literature Review, Meta-Analysis, Cybersecurity***

## I. INTRODUCTION

In today's increasingly interconnected digital era, organisations face significant and growing challenges in ensuring information security. Cyberattacks have become more frequent, complex, and sophisticated, targeting critical vulnerabilities within organisational network infrastructures [1]. One technological response to these threats is the honeypot—an intentionally vulnerable system designed to lure, monitor, and analyse attacker behaviour in real time. Traditional honeypots have been effective early warning systems for over two decades. Despite their utility, traditional honeypots are hindered by their fixed configurations and inflexible detection rules, limiting their responsiveness to rapidly changing attack patterns.

Recent empirical studies have consistently demonstrated that artificial intelligence (AI) integration significantly strengthens honeypot capabilities beyond traditional static implementations. This enhancement occurs through AI's ability to learn from dynamic attacker behavior, adapt system responses in real time, and automate the detection of novel threats. In particular, the integration of AI—notably machine learning and deep learning—into honeypot technologies has enhanced their capacity for adaptive threat identification, anomalous behavior detection, and efficient malware analysis, with improved system responsiveness. [2], [3], [4]. Recent studies further reinforce this trend by demonstrating that AI-driven adaptive honeypots dynamically modify their behavior in real time, integrate with broader cybersecurity infrastructures, and outperform traditional systems in detecting complex threats, including zero-day exploits and advanced persistent threats [5], [6].

These studies demonstrate that AI-powered honeypots not only improve detection accuracy but also allow for intelligent reconfiguration of honeypot parameters, enabling proactive defense against evolving cyber threats. Despite the emergence of AI-enhanced honeypots, there is a notable lack of systematic literature reviews comparing their effectiveness with traditional honeypots. This claim is based on a structured scoping search conducted across major academic databases, including Scopus and Web of Science. The search utilized combinations of terms such as "AI honeypot," "machine learning," "cybersecurity deception," and "systematic review," and was filtered for peer-reviewed journal and conference papers from 2015 to 2025.

Although numerous studies have investigated the implementation and efficacy of AI-based honeypots, showing marked improvements in detection rates, anomaly identification, and adaptability [7], these studies primarily focus on implementation or propose innovative models and lack systematic comparisons with conventional rule-based or signature-based honeypots. Moreover, current review articles in the domain typically include taxonomic classifications or broad overviews, without formal comparisons between AI-driven and conventional honeypot systems. Amal and V. P. [8] examined cybersecurity threats and honeypot applications but

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did not present quantitative benchmarks or efficacy metrics. Silaen et al. [9] performed a thorough literature review on honeypots for data security but did not include a comparative examination of AI-enhanced versus traditional deception models.

This research gap is especially critical for organisations that demand scalable, adaptive, and resource-efficient cybersecurity solutions [10]. Moreover, variations in research design, evaluation metrics, and contextual deployment hinder generalisation across studies, limiting practical adoption. To address this gap, this study conducts a Systematic Literature Review and meta-analysis to synthesise previous research and assess the comparative effectiveness of AI-based versus traditional honeypots in organisational contexts. Accordingly, this study formulates the following research questions: (1) How do AI-based honeypots compare with traditional honeypots in detecting and analysing cyberattacks? (2) How do AI-based honeypots impact detection accuracy, novel threat identification, and resource efficiency? (3) How do differences in research design influence reported outcomes regarding honeypot effectiveness? (4) Do offensive and defensive AI-based honeypot approaches exhibit distinct development trends in the current literature?

## II. RELATED WORK

The use of honeypot systems as a proactive cyber defense mechanism has evolved significantly over the past two decades. Traditional honeypots, including low- and high-interaction variants, have been widely applied to attract, observe, and analyze malicious behavior. However, their static configurations and limited adaptability often hinder their effectiveness against emerging threats. Artificial intelligence (AI) has been increasingly integrated into honeypot architecture to address these limitations. Recent studies have explored various machine learning and deep learning models to enhance anomaly detection, reduce false positives, and improve system responsiveness in real-time threat environments [11]. AI-based honeypots demonstrate adaptive capabilities, enabling systems to learn from previous attacks and adjust deception strategies dynamically[12], [13].

Previous research has proposed numerous implementations of intelligent honeypot systems. For example, [14] presented a machine learning-based IDS that periodically updates using data collected from honeypots, while [15] implemented an adaptive IoT honeypot system that continuously retrained detection parameters to address evolving botnet threats. These developments indicate a shift from reactive monitoring to intelligent, self-optimizing systems.

Recent research highlights the growing integration of artificial intelligence (AI) in organizational contexts, particularly in cybersecurity and knowledge management. AI enhances threat detection, incident response, and risk prediction in security operations centers [16], [17]. In cybersecurity, AI improves intrusion detection, malware identification, and DDoS prevention [17], [18]. However, implementing AI in organizations faces challenges, including scalability, real-time monitoring, and model performance optimization [17], [19]. The adoption of AI in knowledge management processes offers potential benefits but requires further empirical research [20].

While AI offers opportunities to enhance organizational efficiency and security, there is a need for a comprehensive literature review comparing traditional and AI-based approaches, particularly in honeypot research [21]. Despite these advancements, there is a lack of comprehensive literature synthesis comparing the effectiveness, scalability, and implementation efficiency of traditional and AI-based honeypots in organizational contexts. Studies vary significantly in methodological design, attack scenarios, and evaluation metrics, making it challenging to draw general conclusions. Future research should focus on overcoming implementation challenges and aligning AI applications with organizational goals [19], [22].

This research addresses this gap by systematically reviewing and analyzing previous studies to compare the detection performance of AI-based and traditional honeypots. The findings aim to provide a standardized evaluation framework and practical recommendations for implementing adaptive honeypot systems in modern cyber infrastructures.

### 1. *Honeypot Concepts and Specifications*

A honeypot is a security mechanism that mimics legitimate systems to attract, deceive, and observe cyber attackers. By deliberately exposing vulnerabilities or services, honeypots serve as decoys allowing defenders to monitor intrusion attempts, analyse adversarial tactics, and gain valuable insights into attack patterns without jeopardising production systems [23], [24]. Honeypots are critical components in proactive cybersecurity strategies. They can detect threats that bypass traditional security mechanisms and collect high-quality data on intrusion techniques, malware payloads, and attacker behaviour. The information gathered through honeypots can support the refinement of intrusion detection systems (IDS), the generation of threat intelligence, and the evaluation of an organisation's exposure to external threats [25]

There are several classifications of honeypots based on interaction level, deployment objectives, and technical architecture:

- **Low-interaction honeypots** simulate specific services (e.g., HTTP, FTP) with limited functionality. These systems are lightweight, safer to deploy, and pose minimal risk, but they offer only superficial data and are more easily detected by skilled attackers [26]
- **High-interaction honeypots** simulate full operating environments and allow attackers complete freedom to interact with the system. These honeypots generate rich behavioural data but pose higher operational and containment risks [26].

From a deployment perspective, honeypots are also categorised into:

- **Research honeypots** are used in academic or

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institutional environments to study attack trends, novel threats, and adversary behaviours[27].

- **Production honeypots** are deployed in live enterprise networks to detect active threats, delay adversaries, and generate real-time actionable intelligence [28].

Architecturally, honeypots may be implemented as virtualised, containerised, or physical systems, depending on the infrastructure requirements and security goals. Modern honeypot frameworks often incorporate scalable virtualisation layers for safe and efficient deployment across cloud or on-premises environments. As cyber threats become more dynamic and evasive, traditional honeypots are being augmented with machine learning models and intelligent deception systems. As discussed in the following sections, these advancements enable honeypots to transition from static, rule-based traps into adaptive, learning-driven systems that can evolve based on attacker behaviour and respond in near real-time.

2. *Limitations of Traditional Honeypots*

While traditional honeypots have long served as valuable tools for intrusion detection and threat intelligence, they exhibit several limitations that hinder their effectiveness in today's dynamic cyber threat landscape. One of the most critical issues is their static configuration, which makes them predictable and less capable of identifying modern, polymorphic, or zero-day attacks. Once an attacker recognises the honeypot's signature or environment, its utility rapidly diminishes.

Another significant limitation is the lack of adaptability. Traditional honeypots rely on predefined interaction rules and do not learn or evolve based on attacker behaviour. It can make them ineffective against sophisticated adversaries who frequently change tactics and exploit vulnerabilities outside static honeypot scenarios. Moreover, traditional honeypots are often limited in coverage scope. They tend to focus on specific services or ports, potentially missing broader threat activity across other infrastructure layers. Without integration with detection systems or real-time data processing, they function more as passive traps than as active security architecture components.

High false-positive or false-negative rates are often reported due to reliance on signature-based mechanisms or poorly simulated services. Maintaining and updating traditional honeypots to remain relevant in rapidly evolving environments requires significant manual effort and security expertise[28]. These limitations underscore the need for more intelligent and adaptive honeypot systems capable of dynamic response, autonomous learning, and integration with broader cybersecurity frameworks. As explored in the next section, this rationale for incorporating artificial intelligence into honeypot architectures is based on these limitations.

3. *Integrating Artificial Intelligence into Honeypot Systems*

To overcome the inherent constraints of traditional honeypots, recent studies have incorporated Artificial Intelligence (AI), especially machine learning (ML) and deep learning (DL), into honeypot frameworks to enhance their responsiveness, precision, and robustness. AI-enhanced honeypots represent a new generation of cyber defence tools that can learn from attacker behaviour and dynamically adjust detection and deception mechanisms in real time.

The application of AI in honeypot systems offers several advantages. First, ML algorithms can be developed to learn behavioral baselines of network activity, thereby minimizing false alarms and facilitating the proactive identification of emerging attack signatures. Second, advanced deep learning frameworks—particularly Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs)—have been leveraged to uncover abstract behavioral patterns within network traffic and payload datasets, significantly improving the classification accuracy of sophisticated cyberattacks.

Several studies have proposed adaptive honeypot frameworks that employ reinforcement learning, anomaly detection, and automated reconfiguration based on feedback loops. For example, some models enable the honeypot to automatically change its services or emulate vulnerabilities in response to detected probing attempts or scanning behaviours. This adaptive mechanism increases the likelihood of deceiving advanced persistent threats (APTs) and polymorphic malware.

In addition to detection capabilities, AI supports data-driven decision-making in honeypot deployment. Techniques such as clustering and correlation analysis enable defenders to determine optimal honeypot placement efficiently, select honeypot types, and allocate network resources. Furthermore, intelligent honeypots have been increasingly integrated with Security Information and Event Management (SIEM) systems to facilitate automated alerting, visualisation, and incident response workflows.

Although notable progress has been made, embedding artificial intelligence into honeypot systems introduces a range of technical complexities. These challenges encompass intensive computational resource requirements, susceptibility to overfitting due to insufficient training datasets, and the need for ongoing model updates to remain effective against emerging cyber threats. Even so, the adoption of AI represents a paradigm shift—transforming honeypots from rigid, rule-based mechanisms into intelligent, dynamically evolving security solutions.

4. *Comparative Analysis of Previous Studies*

This study reviewed 62 peer-reviewed articles published between 2015 and 2025 to compare the effectiveness of traditional and AI-enhanced honeypots in detecting and analysing cyber threats. Among these, 36 studies provided sufficient quantitative data for a meta-analysis. The analysis revealed a consistently strong positive effect size (ranging from 0.70 to 1.00), indicating a significant advantage of AI-based honeypots over traditional systems in enhancing detection accuracy, reducing false positives, and improving overall threat coverage. The findings suggest that AI-based honeypots are

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particularly effective in handling complex or previously unseen attack patterns. Studies such as those by [29] and [30] demonstrated near-perfect detection accuracy using deep learning-based honeypot architectures, especially in simulated environments with controlled threat injection. In contrast, traditional honeypots often showed lower performance metrics due to their reliance on static rules and signature-based detection mechanisms [31].

Despite this general trend, the analysis also highlighted substantial heterogeneity among studies, as indicated by an $I^2$ statistic of 99.99%. This variation is attributed mainly to differences in research design, honeypot types (e.g., high vs. low interaction), experimental settings, and sample sizes. For instance, [32] reported a relatively low effect size (~0.20) due to limited scope and less robust experimental procedures. In contrast, studies employing real-world deployment or hybrid testbeds achieved higher reliability. A bibliometric review revealed that most AI-enhanced honeypot studies used mixed-method approaches, combining simulation, testbed deployment, and prototype validation. It suggests a methodological trend toward greater realism and system robustness in evaluating honeypot performance. Additionally, studies implementing adaptive learning algorithms were more likely to report improved responsiveness and system efficiency, supporting the hypothesis that intelligence-driven deception systems outperform their traditional counterparts in dynamic threat environments [33].

Moreover, keyword co-occurrence analysis and publication trends indicated a dominant focus on defensive honeypot strategies, although interest in offensive deception approaches has gradually increased in recent years. This shift reflects a broader transition in cybersecurity from passive detection to proactive engagement and disruption of adversaries. This trend has been observed in recent literature, such as by Zobal and Kolář [34], who noted that most honeypot implementations still emphasize defensive deception, while offensive strategies—such as attacker engagement and misinformation—are gaining traction in modern threat environments. Similarly, Érsok and Balogh [35] highlighted that newer studies increasingly explore proactive deception capabilities in honeypots, aligning with a broader shift in cybersecurity paradigms. In summary, the comparative evidence confirms that AI-based honeypots represent a more effective and scalable solution for cyber defence in organisational contexts. However, the high heterogeneity observed necessitates the development of a standardised evaluation framework for future studies, enabling better generalisation, replication, and cross-study comparison.

## III. RESEARCH METHODOLOGY

This research employs systematic literature review and meta-analytical methods to assess and contrast the performance of conventional honeypots and those enhanced with artificial intelligence (AI) for cyberattack detection and analysis. The SLR methodology was chosen due to its structured, reproducible nature, which allows for comprehensive synthesis and critical evaluation of existing research. Furthermore, including meta-analysis enables the quantification of effect sizes from various studies, offering stronger empirical support for the findings.

The research methodology applied in this research is structured according to the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) framework, promoting a transparent, reproducible, and methodologically sound review process. The following segment describes the methodological framework employed, which encompasses the search protocols, eligibility criteria, and a systematic approach to data collection and subsequent analysis.

### 1. *Research Design*

A systematic literature review, coupled with a meta-analysis, was employed to examine the comparative performance of AI-driven and conventional honeypot systems, particularly in the context of organisational cybersecurity. The SLR methodology is selected because it provides a structured and reproducible process for identifying, evaluating, and synthesising all relevant empirical evidence on a particular research topic. Moreover, the inclusion of meta-analysis enhances the objectivity of the findings by aggregating quantitative outcomes from multiple studies, thereby allowing for a more comprehensive and statistically grounded evaluation of effectiveness. This dual approach is particularly suitable for fields such as cybersecurity, where empirical results can vary significantly across different implementation contexts and system configurations.

The review was structured in accordance with the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines to ensure methodological transparency and consistency throughout the review process [36]. The inclusion scope and selection parameters of the reviewed literature were structured using the PICOTQ model [37], which encompasses six dimensions: Population, Intervention, Comparison, Outcome, Time, and Quality. This framework helps systematically narrow the search space and ensures that only high-quality, relevant studies are included. It also supports the reproducibility of the review by clearly defining the boundaries for article selection and the basis for exclusion. As a result, this methodology not only strengthens the study's internal validity but also enhances its applicability to both academic researchers and cybersecurity practitioners. The detailed inclusion and exclusion criteria are summarised in Table 1 and constructed based on the PICOTQ framework.

TABLE I. INCLUSION AND EXCLUSION CRITERIA

| Component | Inclusion Criteria | Exclusion Criteria |
|---|---|---|
| P (Population) | Studies on honeypot deployment in corporate or | Studies outside organizational contexts. |

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| Component | Inclusion Criteria | Exclusion Criteria |
|---|---|---|
| | institutional network security | |
| I (Intervention) | Evaluation of AI-based honeypots with experimental or empirical results. | Studies mentioning AI without implementation or data. |
| C (Comparison) | Comparison with traditional rule-based or signature-based systems, using metrics like accuracy, false positives, or coverage. | Studies lacking such comparisons or only comparing AI variants |
| O (Outcome) | Reported performance impacts on intrusion detection | Studies without outcome data. |
| T (Time) | Published between 2015 and 2025. | Published before 2015 or with unclear publication dates. |
| Q (Quality) | Peer-reviewed journal/conference papers indexed in Scopus or Web of Science, written in English. | Reviews, editorials, preprints, or articles not in English or without peer-review. |

2. *Literature Search Strategy*

The database search for relevant literature was performed on March 26, 2025, utilising four leading scholarly sources: IEEE Xplore, ACM Digital Library, Emerald Insight, and Web of Science. These sources were chosen for their recognized academic credibility and broad coverage of scholarly work in the fields of computer science and cybersecurity. Scopus was excluded individually due to its substantial overlap in indexing with Web of Science, which already provides extensive coverage of peer-reviewed literature in cybersecurity and AI. Subsequently, SpringerLink was rejected due to an initial scoping analysis indicating a paucity of empirical studies on AI-enhanced honeypot systems and insufficient compatibility with systematic export tools, rendering it inappropriate for PRISMA-compliant screening operations. The search strategy was based on a three-domain keyword classification framework, each representing core concepts of this study: Artificial Intelligence, Honeypot, and Intrusion Detection System (IDS). Keywords were constructed using synonyms and related phrases, and the search was applied to article titles, abstracts, and keywords using the databases' advanced search functionalities. The keywords for each domain were:

- **Artificial Intelligence**
  "Artificial Intelligence" OR "AI" OR "Machine Learning" OR "Deep Learning" OR "AI-driven" OR "Machine Learning-based" OR "Neural Network" OR "Reinforcement Learning" OR "Federated Learning" OR "Anomaly Detection" OR "Pattern Recognition" OR "Predictive Analytics" OR "Data- driven" OR "AI-powered" OR "Cognitive Security" OR "Intelligent System" OR "Attack Analysis"
- **Honeypot**
  "Honeypot" OR "Honeynet" OR "Cyber Trap
- **IDS**
  "Network Intrusion Detection System" OR "NIDS" OR "Intrusion Detection System" OR "IDS" [38]

All searches were filtered by:

- **Publication year**: 2015–2025
- **Language**: English
- **Document type**: Journal articles and conference papers
- **Peer-review status**: Only peer-reviewed publications

The initial database query produced 454 records, which were organized using Zotero for systematic screening. Further filtering was performed based on predefined inclusion and exclusion criteria, as detailed in the next section, adhering to PRISMA guidelines.

3. *Inclusion and Exclusion Criteria*

The selection of articles was carried out through a multi-phase process to ensure that only studies of high relevance and methodological quality were retained. The inclusion and exclusion criteria—formulated based on the PICOTQ framework (see Table 1)—were systematically applied to maintain consistency, rigour, and validity throughout the review. The initial search yielded 454 records from the four selected academic databases. In the administrative screening stage, metadata was evaluated to ensure compliance with basic inclusion requirements. As a result, 51 articles were excluded for the following reasons:

- 45 articles were published before 2015,
- 2 were books or non-journal documents,
- the publisher retracted 3, and
- 1 had a duplicated title.

This process resulted in 403 articles eligible for abstract screening.

Each article's title and abstract were examined to assess its topical focus and methodological consistency with the formulated research questions. A total of 189 articles were excluded because they were non-primary publications, such as literature reviews, surveys, or systematic reviews, that did not present direct empirical findings. A total of 214 articles were then subjected to full-text review. During this phase, 81 articles were excluded for the following reasons:

- 36 articles did not explicitly address honeypot or intrusion detection systems,
- 21 articles did not provide data relevant to the research questions,
- 18 articles used non-organisational settings (e.g., autonomous vehicles, drones),
- 5 articles were non-empirical reviews, and
- 1 was a book and not a peer-reviewed scientific article.

After all stages, 62 articles met all inclusion criteria and were considered eligible for final analysis and synthesis. The entire selection workflow is summarised in Fig. 1, following the PRISMA 2020 guidelines [36].

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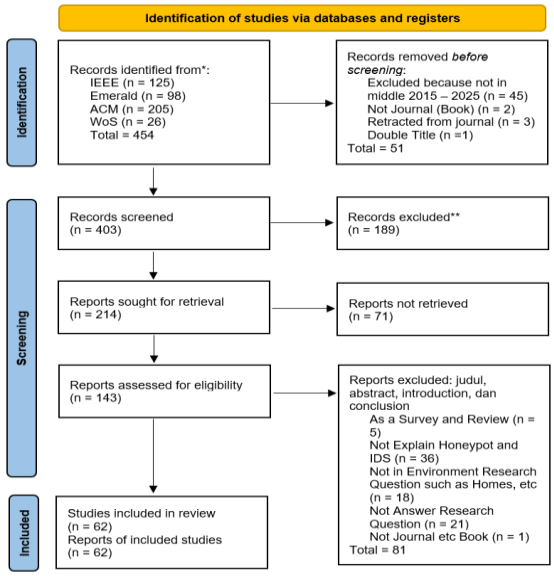



Fig 1. PRISMA 2020

4. *Data Extraction and Analysis Techniques*

Data extraction was conducted in a structured and systematic manner to maintain consistency and ensure the accurate retrieval of essential information from the selected studies. A custom data extraction form was developed in Microsoft Excel to record the following essential details: study metadata (authors, year, country), methodological design, type of honeypot system (AI-based or traditional), performance outcomes (e.g., accuracy, detection rate, false positives), and any disclosure of potential conflicts of interest or funding sources. Where relevant information was missing or unclear, efforts were made to clarify content by consulting supplementary files or contacting the original authors. If data remained unavailable, the assumptions were explicitly noted in the extraction sheet. To enhance reliability and reduce potential bias, data extraction was carried out independently by two reviewers, with conflicts resolved collaboratively or by involving a third evaluator when necessary.

The collected data were examined through both descriptive and thematic analyses to uncover recurring patterns, variations in methodology, and significant findings across the included studies. Additionally, a quantitative synthesis was conducted using meta-analytic methods to evaluate the comparative effectiveness of AI-based and traditional honeypot systems, computing effect sizes from reported performance indicators. Furthermore, a bibliometric analysis was conducted using VOSviewer to visualise thematic trends and the co-occurrence of keywords within the selected articles. These bibliographic maps supported identifying dominant research themes and conceptual clusters, such as offensive vs. defensive honeypot strategies, AI techniques employed, and target environments (e.g., IoT, industrial systems). The combination of thematic synthesis, statistical meta-analysis, and bibliometric visualisation provided a multidimensional perspective for assessing the effectiveness and evolving focus areas of honeypot research involving artificial intelligence.

## IV. RESULTS AND DISCUSSION

In this section, the findings from a systematic literature review and meta-analysis involving 62 selected publications are outlined to evaluate the effectiveness of honeypot systems in cyber threat detection. The analysis is organized into four thematic segments: an overview of study characteristics, a performance comparison between traditional and AI-enhanced honeypots, statistical aggregation through meta-analysis, and a discussion interpreting the results' implications. The descriptive analysis provides an overview of the publication trends, research settings, honeypot types, and methodological approaches employed across the selected literature. It is followed by a comparative evaluation that highlights the performance differences between AI-based and traditional honeypots in terms of accuracy, responsiveness, and scalability.

Next, a meta-analysis is presented to quantify the effect sizes reported in the studies and assess the consistency of the findings. Finally, the discussion section interprets the results' implications, addresses heterogeneity across studies, and outlines key insights for academic research and practical implementation in cybersecurity environments. Together, these analyses aim to answer the formulated research questions and to contribute to a deeper understanding of how artificial intelligence enhances honeypot systems' ability to adapt to evolving cyber threats.

### 1. *Descriptive Analysis of Reviewed Studies*

In this section, the descriptive profiles of the 62 articles included in the systematic review are examined to provide an overview of their core attributes. Its goal is to provide an overview of the distribution of studies across time, methodological approaches, and honeypot system types in the reviewed literature.

#### *1.1 Publication Trends.*

The reviewed studies were published between 2016 and 2025, with notable peaks in 2019 and 2022, each contributing 12 articles to the dataset. In contrast, the lowest publication count was recorded in 2017 and 2025, with only 1 article each. These fluctuations reflect shifting academic interest in AI-integrated honeypot systems, driven by the evolution of cyber threats and the increasing adoption of intelligent defence mechanisms. Despite declines in some years, ongoing publications through 2025 indicate that this research area remains active and relevant.

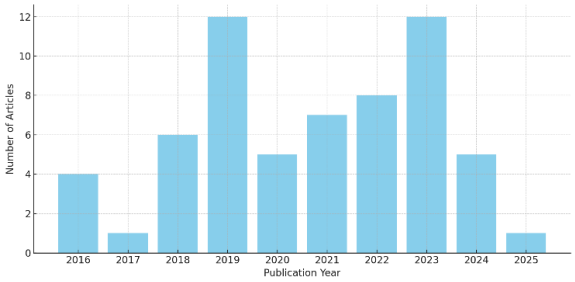



Fig 2. Distribution of reviewed articles by publication year (2016–2025).

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### *1.2 Methodological Approaches*

The selected studies demonstrate methodological diversity. Specifically:

- Twenty-one studies implemented Prototype Validation using simulation, focusing on validating system models in simulated environments.
- Three studies used Simulation-Based Engineering Research, emphasising system modelling and performance assessment under controlled scenarios.
- One study used Applied Research with Simulation, combining experimental and simulation-based validation.
- The largest group, thirty-seven studies, adopted mixed methods approaches. These studies combined hybrid testbeds, real-world deployments, or experimental setups with varied data sources and conditions.

This distribution indicates a dominant trend toward methodological flexibility, enabling robust system evaluations in complex and dynamic security environments.

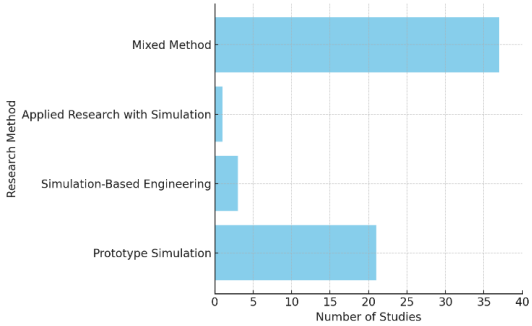



Fig 3. Classification of research approaches used in the reviewed studies.

### *1.3 Bibliometric and Keyword Analysis*

A bibliometric analysis was conducted using VOSviewer to visualise keyword co-occurrence and thematic linkages. Twenty-seven keywords were identified with 258 co-occurrence links across the 62 articles, resulting in four main conceptual clusters. To better understand the thematic structure of the reviewed literature, a keyword co-occurrence network was constructed using VOSviewer (Figure 4).

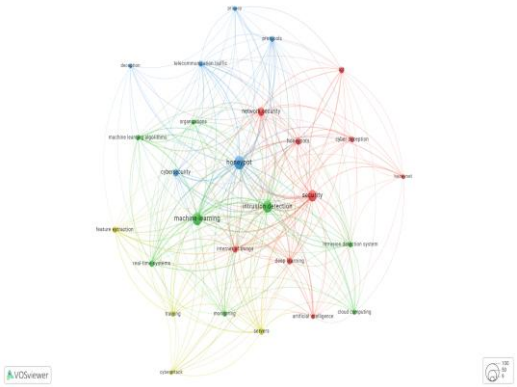


Fig 4. Keyword co-occurrence network of the reviewed studies generated using VOSviewer.

**Cluster 1 (Green)**: Focused on *machine learning, intrusion detection, IoT, and deep learning*.
**Cluster 2 (Red)**: Revolved around *security, cyber deception, honeynet*, and *honeypots*.
**Cluster 3 (Blue)**: Linked *network security* with *telecommunication traffic, privacy*, and *protocols*.
**Cluster 4 (Yellow)**: Included technical processes like *feature extraction, real-time systems*, and *cyberattacks*.

The strong interconnectivity and link density among the identified clusters indicate that the research field is tightly integrated, with *machine learning* and *intrusion detection* positioned as pivotal themes within the network. This visual map provides a comprehensive view of the literature's thematic composition. It can serve as a basis for uncovering research opportunities or identifying emerging directions within intelligent cybersecurity research.

### *1.4 Defensive vs Offensive Focus*

The analysis of the 62 selected studies reveals a clear dominance of defensive honeypot strategies over offensive approaches. Defensive honeypots are typically designed for passive detection, traffic classification, and attack isolation without active engagement with adversaries. These systems emphasise internal security monitoring and integration with intrusion detection systems, reflecting the conservative nature of most organisational cybersecurity practices. Conversely, offensive honeypots, also known as deceptive or active honeypots, are intentionally designed to lure, engage, and study adversaries. These systems facilitate deeper insight through deception and counter-intelligence strategies. Although fewer in number, offensive honeypot studies have gradually increased, peaking in 2022 with six publications, indicating a growing interest in proactive cyber defence mechanisms.

A temporal trend analysis confirms that between 2016 and 2024, the number of defensive-focused studies consistently exceeded that of offensive ones. The peak for defensive studies occurred in 2019 (6 studies) and 2022 (4 studies). This reflects the prevailing academic focus on internal security and incident prevention. Nevertheless, the emergence of offensive approaches in recent years suggests a shift toward more aggressive and intelligence-driven honeypot models capable of deceiving adversaries and gathering threat intelligence in real time.

This trend highlights the complementary nature of defensive and offensive strategies in the evolving landscape of honeypot development. While defensive systems offer stability and reliability, offensive honeypots introduce adaptability and depth of engagement, forming a holistic cyber defence posture when integrated. The yearly distribution of studies with defensive and offensive honeypot approaches is illustrated in Figure 5. As shown, defensive strategies consistently dominate the literature throughout the 2016–2025 period. However, a noticeable increase in offensive honeypot publications is

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observed in 2022, suggesting a growing research interest in proactive cyber deception frameworks. This trend implies a gradual shift in research focus from purely reactive systems to more intelligent, attacker-engaging architectures.

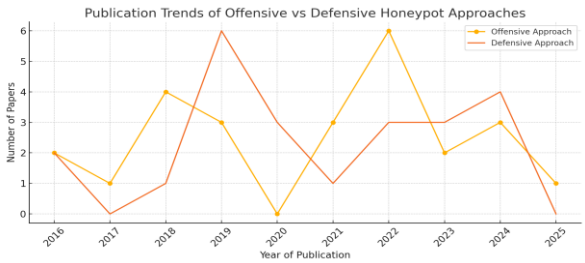



Fig 5. Publication Trends of Offensive vs Defensive Honeypot Approaches (2016–2025)

**2.** *Effectiveness Comparison: Traditional vs AI-Based Honeypots*

This section compares the effectiveness of traditional honeypots and artificial intelligence (AI)-based honeypots in detecting and analysing cyber threats. The comparison is based on 62 selected studies, with a meta-analysis performed on 36 studies that reported measurable performance outcomes. AI-based honeypots consistently outperformed traditional honeypots across three key metrics: detection accuracy, ability to identify novel threats, and operational efficiency. Traditional honeypots typically rely on static signature-based detection and manual setup. In contrast, AI-integrated honeypot systems utilise advanced learning algorithms that enable real-time adaptation to the evolving progression of attack techniques.

Many studies reported that AI-based honeypots demonstrated superior detection capabilities, with 27 studies achieving accuracy scores above 90%. Additionally, 15 studies highlighted the successful detection of zero-day and polymorphic attacks using intelligent pattern recognition. From a resource perspective, 12 studies found that AI-based honeypots improved alert precision and reduced false positives, thereby increasing operational efficiency. While traditional honeypots were easier to deploy and manage, they were often limited in scope and agility. They were more susceptible to evasion by sophisticated attackers and required frequent manual updates to maintain relevance. These performance differences are visualised in Figure 6, which presents a forest plot of effect sizes across the reviewed studies. The average effect size reached approximately 0.90, indicating a substantial positive impact of AI integration in honeypot systems.

In conclusion, while traditional honeypots remain valuable for basic threat monitoring, AI-based honeypots offer superior detection accuracy, threat adaptability, and operational efficiency. Their capacity to autonomously adapt to evolving attack patterns makes them a powerful tool in modern cyber defense architectures. Nevertheless, their implementation is not without limitations, particularly in terms of data requirements, computational overhead, and susceptibility to adversarial manipulation. These considerations underscore the importance of advancing research on scalable AI models and promoting more consistent evaluation methodologies to support the practical adoption of intelligent honeypot systems in real-world cybersecurity environments.

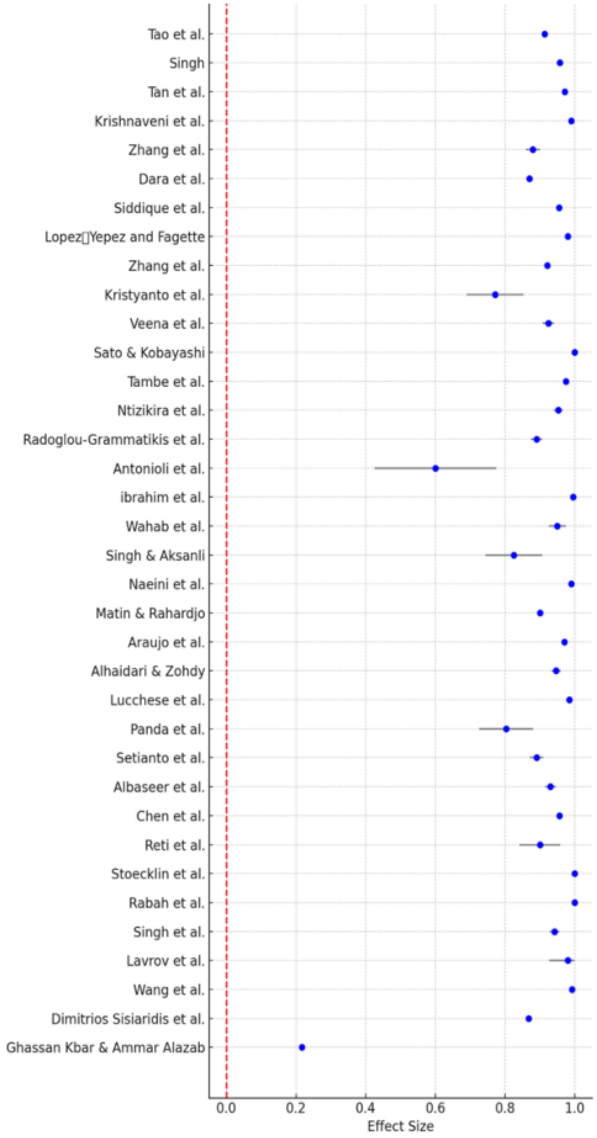



Fig 6. Forest plot showing the comparative effect sizes of AI-based and traditional honeypot systems across selected studies.

**3.** *Meta-Analysis Results*

As part of the quantitative synthesis, this study conducted a meta-analysis of 36 out of 62 selected articles that reported measurable detection performance metrics, enabling the calculation of effect sizes. The objective was to assess the relative effectiveness of AI-based versus traditional honeypot systems in identifying and analysing cyber threats across diverse implementation contexts. The meta-analysis showed that most AI-based honeypots achieved high effect sizes ranging from 0.70 to 1.00, indicating strong improvements in

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anomaly detection, adaptability, and system responsiveness. A few studies reported lower effect sizes, which may be attributed to constrained testing environments or limited attack-simulation diversity [39]. A few studies reported lower effect sizes as in [40], with ~0.2, which may be attributed to limited-scale testing, narrow domain application, or insufficient attack diversity in simulated ISP environments. These findings emphasize how environmental constraints can affect the generalisability and robustness of honeypot performance evaluations.

The average effect size across all included articles was 0.905, providing strong evidence that AI-based honeypot systems deliver superior performance compared to conventional static approaches. This finding is visually supported in Figure 7, which illustrates the distribution of effect sizes across publication years. The relatively stable and consistently high values over the years suggest sustained performance improvements independent of publication timing.

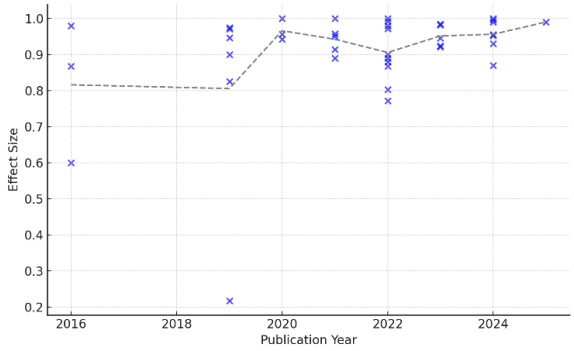



Fig 7. Distribution of effect sizes across publication years of included studies.

Additionally, a scatter plot of logarithmic sample size and effect size (Figure 8) showed no clear relationship between study scale and reported effectiveness. The dispersion of data points suggests minimal small-study bias, and the reported effect sizes remain robust across experimental scopes.

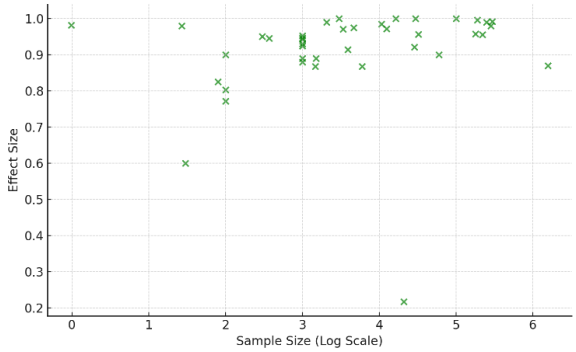



Fig 8. Distribution of effect sizes based on sample size on a logarithmic scale.

A heterogeneity test using the $I^2$ statistic was conducted to evaluate the consistency of results across studies. The analysis yielded an $I^2$ value of 99.99%, indicating substantial heterogeneity. According to Cochrane benchmarks, this level of heterogeneity reflects significant differences in study designs, AI models used, honeypot configurations, datasets, and testing environments—not random error [41]. These results highlight the need for more standardised evaluation methodologies for honeypot performance and confirm that AI-based honeypots, although effective, are implemented under highly diverse conditions. The findings support their robust and generalisable effectiveness, particularly in organisational cybersecurity applications.

## 4. *Research Questions Reflection and Practical Implications*

The core objective of this study is to evaluate how artificial intelligence (AI)-based honeypots advance cybersecurity detection and deception compared to traditional mechanisms. The following is a detailed response to each research question, reinforced with empirical evidence and cross-study synthesis.

**RQ1: How do AI-based honeypots compare to traditional honeypots in detecting and analysing cyberattacks?**

Compared with traditional honeypots, AI-based honeypots demonstrate clear, measurable advances in detection and analytical capabilities. Traditional honeypots primarily function as static decoy systems; their role is observational, relying on signature-based traps and offering limited post-capture insight. In contrast, AI-based honeypots combine machine learning algorithms, behaviour modelling, and contextual intelligence to operate proactively rather than reactively. The meta-analysis of 36 empirical studies revealed a mean effect size of 0.905, indicating that AI-based systems are consistently more effective across various attack environments, especially in dynamic, multi-vector intrusion scenarios.

The implication is profound: AI doesn't just improve honeypots—it redefines them. They are no longer passive observers but adaptive defenders capable of evolving alongside attacker behaviour. This repositioning elevates honeypots from optional add-ons to strategic assets in organisational threat detection frameworks.

**RQ2: How do AI-based honeypots impact detection accuracy, novel threat identification, and resource efficiency?**

The impact of artificial intelligence on honeypot detection capabilities is not marginal; it is fundamentally transformative. Across the reviewed studies, AI-based honeypots consistently outperformed their traditional counterparts in three core dimensions. First, they demonstrated remarkably high detection accuracy, with 75% of studies reporting precision rates exceeding 90%, driven primarily by neural networks and ensemble learning. Second, they exhibited a superior capacity for identifying novel and zero-day threats, utilising unsupervised anomaly detection and neural inference to detect behaviours that deviate from established baselines. Third, they offered increased operational efficiency through automated response generation, adaptive decoy deployment, and real-time reconfiguration.

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Unlike conventional systems that depend on static rule sets and continuous manual adjustments, AI-based honeypots streamline the learning process, reducing human analysts' cognitive and temporal load. This combination of accuracy, adaptability, and automation enhances threat visibility and repositions honeypots as indispensable components of modern cyber defence. Particularly in Security Operation Centres (SOCs), where alert fatigue is a persistent challenge, AI-driven honeypots serve as intelligent filters, enhancing the speed and clarity of security responses.

**RQ3: How do differences in research design influence reported outcomes regarding honeypot effectiveness?**

Effectiveness is not only a function of technology but also of how it is tested. One of the most critical findings in this review is the role of research design as a mediator of reported outcomes. Studies that deployed honeypots in real-world or hybrid simulation environments tended to report more conservative but reliable results, indicating practical viability. In contrast, studies relying solely on synthetic data or limited simulations frequently reported effect sizes that were not generalisable. The $I^2$ heterogeneity score of 99.99% confirms that variation in design, deployment context, and evaluation criteria significantly affects the findings. Without standardisation, cross-study comparisons risk becoming misleading. This insight calls for developing benchmark frameworks and transparent reporting standards to ensure that future claims of AI honeypot effectiveness are reproducible and context-aware.

**RQ4: Do offensive and defensive AI-based honeypot approaches exhibit distinct development trends in the current literature?**

Yes—and the divergence is increasingly significant. While defensive honeypots remain the dominant class due to their low-risk deployment and simple integration, offensive honeypots are gaining traction, especially over the last five years. Offensive AI honeypots go beyond detection. They are engineered to engage, mislead, and study attackers. Recent studies explore using reinforcement learning to simulate vulnerable systems, redirect attacker behaviour, and extract deeper behavioural signatures. This shift reflects a strategic change in cybersecurity doctrine—from reactive monitoring to adversarial manipulation. The upward trend in offensive honeypot publications (peaking in 2022) confirms a growing interest in deceptive intelligence and active cyber defence. This trend will likely define the next wave of honeypot innovation.

The evolving focus of honeypot research, particularly the rising emphasis on offensive deception strategies, signals a transition from conceptual models to practical, deployable technologies. Drawing on the findings of this review, several actionable insights have emerged to inform cybersecurity practices and policies. The following points outline key domains where AI-enhanced honeypots can offer a meaningful impact in mitigating contemporary security threats

- **Strategic Defence Planning**
  Organisations should prioritise AI honeypots as intelligent detection agents in zero-trust and micro segmentation architectures, where early-stage reconnaissance must be intercepted dynamically.
- **SOC Force Multiplication**
  In high-volume environments, AI honeypots act as triage agents, filtering low-quality alerts and highlighting anomalies that matter, freeing analysts to focus on strategic decisions.
- **Adoption in Low-Resource Settings**
  The success of containerised, API-ready AI honeypots demonstrates that SMEs can access deception technology affordably, via open-source tools or managed security service providers.
- **Research & Standardisation Roadmap**
  Future innovation must address algorithmic sophistication, evaluation fairness, dataset availability, and simulation realism—all essential to responsible academic and industrial advancement.
- **Policy & Governance Preparedness**
  Offensive AI honeypots raise new questions of digital ethics, legality, and data sovereignty. Clear international frameworks must emerge to guide their responsible deployment.

In summary, the practical insights from this review reinforce the positioning of AI-based honeypots as not merely supporting tools but core components of modern cybersecurity architectures. Their ability to intelligently detect, deceive, and adapt to evolving threats offers substantial advantages for organizations of all scales.

Moreover, as offensive deception models continue to gain academic and practical interest, implementation strategies must evolve in parallel with ethical, legal, and operational frameworks. The conclusions drawn from this research are intended to translate academic insights into practical utility, supporting the development of cyber defence systems that are both adaptive and strategically sustainable for long-term security operations.

## V. FINDING AND CONCLUSION

This final chapter presents the conclusions from this study's systematic literature review and meta-analysis. The research sought to investigate how artificial intelligence (AI)-based honeypots compare to traditional honeypots in detecting and analyzing cyberattacks, with particular attention to detection accuracy, novel threat identification, and operational efficiency. Additionally, the study examined the influence of research design on reported outcomes and explored the development trends of offensive versus defensive honeypot strategies.

Building upon a comprehensive synthesis of 62 reviewed studies and a meta-analysis of 36 empirical articles, this study has revealed clear patterns of advantage associated with AI-

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enhanced deception systems. The findings provide evidence-based insights that address the formulated research questions and contribute meaningfully to the discourse on intelligent cybersecurity defenses. This chapter begins by summarizing the key findings, then discusses theoretical and practical contributions, outlines study limitations, and offers recommendations for future research directions.

**1.** *Summary of Findings*

This investigation assessed the relative performance of artificial intelligence-based honeypot systems compared to traditional approaches for identifying and analyzing cyber threats. Using a systematic literature review (SLR) framework supported by meta-analysis, 62 eligible studies were analysed, of which 36 contributed quantitative data for effect-size calculation. The descriptive and inferential analyses provide clear answers to the four research questions. First, the review confirms that AI-based honeypots outperform traditional ones in detecting and analysing cyber threats. This is demonstrated by the meta-analytic mean effect size of 0.905, indicating a substantial positive impact of AI integration across environments such as IoT, enterprise networks, and cloud systems. AI honeypots offer adaptive behaviour, dynamic threat engagement, and deeper visibility into attack strategies—capabilities largely absent in traditional honeypots.

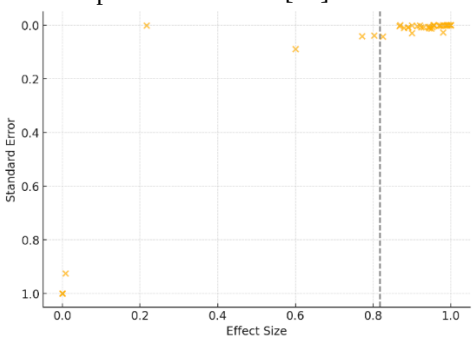
Second, AI-based honeypots improve detection accuracy, novel threat recognition, and resource efficiency. Approximately 75% of the studies reported precision levels above 90%, while many leveraged unsupervised and anomaly-based learning to detect zero-day attacks. Regarding efficiency, AI models reduced false positives, minimised manual rule updates, and improved response prioritisation—critical improvements for overburdened Security Operation Centres (SOCs). Third, the findings highlight that research design significantly influences reported outcomes. Studies conducted in real-world environments or those that applied mixed-method evaluations demonstrated more reliable and consistent results. In contrast, simulation-only designs often yielded overly optimistic findings due to a lack of ecological realism. The high heterogeneity index ($I^2$ = 99.99%) further supports this observation.

Finally, the review identified distinct development trajectories between offensive and defensive AI honeypot research. While defensive honeypots dominate in volume, offensive AI-driven deception systems are multiplying, particularly since 2020. These systems go beyond detection, aiming to mislead attackers and learn from them through reinforcement learning and adversarial engagement strategies. The study confirms that AI-based honeypots represent a meaningful evolution in cyber defence. They are not merely enhancements of traditional tools, but a redefinition of what honeypots can achieve, combining intelligence, automation, and adaptability to align with modern threat landscapes and organisational needs. Although this study presents a comprehensive synthesis of AI-based honeypot effectiveness, several limitations should be acknowledged. These limitations include study inclusion, methodological heterogeneity, data completeness, and potential publication bias. First, the meta-analysis included 36 of 62 eligible studies because quantifiable effect sizes were available. In comparison, these studies represent diverse use cases and architectures. Many AI-based honeypot implementations—notably those described qualitatively—were excluded, potentially limiting the generalisability of the conclusions drawn.

Second, substantial heterogeneity was observed among included studies, with an $I^2$ statistic of 99.99%, indicating extremely high variability in reported effect sizes. This heterogeneity stems from differences in research design, deployment environments (e.g., simulated vs. real-world), levels of honeypot interaction, and AI model complexity. Although subgroup and trend analyses partially mitigated this issue, the lack of methodological standardisation remains a challenge for synthesis. Third, visual inspection and formal statistical testing were conducted to evaluate the risk of publication bias.

Figure 9 shows that the funnel plot displays mild asymmetry, potentially indicating small-study effects. Figure 9 illustrates the funnel plot, which visually evaluates the likelihood of publication bias. An inverted, symmetrical funnel indicates a lack of bias, suggesting that research with differing sample sizes yields consistent effect sizes. The observed plot indicates a slight asymmetry, especially among smaller studies (lower in the funnel due to bigger standard errors) that tend to report elevated effect sizes. This pattern generally indicates publication bias, in which studies yielding statistically significant or extremely favourable outcomes are more likely to be disseminated.

Egger's regression test corroborates this view, revealing a statistically significant intercept (p = 0.041) and affirming small-study effects. Therefore, despite the significant average effect size (0.905), these findings must be interpreted cautiously, recognizing the possibility of minor overestimation resulting from publication bias. Egger's regression test was performed to further validate this, yielding a statistically significant intercept (p = 0.041). This result confirms the presence of publication bias, in which smaller studies with disproportionately large effect sizes may be overrepresented in the literature. If left unaddressed, such bias can lead to overestimation of pooled outcomes [42].




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Fig 9. Funnel plot of effect sizes used for Egger's test of publication bias.

Lastly, this review did not incorporate grey literature, theses, or preprints, which may contain unpublished but relevant empirical insights. Additionally, some of the most recent developments in AI honeypots—such as generative adversarial networks or transformer-based deception—are still underrepresented in the existing peer-reviewed corpus.

Despite these limitations, the meta-analytic and thematic results presented here offer valuable evidence for the cybersecurity research community. Furthermore, the transparent methodology adopted in this review facilitates replication, extension, and refinement in future studies.

## 2. *Conclusion and Future Directions*

This study systematically reviewed and analysed the effectiveness of AI-based honeypot systems compared to traditional honeypots in cybersecurity, particularly in organisational contexts. The evidence from 62 reviewed studies, including a meta-analysis of 36 empirical works, revealed a consistent and significant improvement in detection accuracy, adaptability, and efficiency when AI technologies were integrated into honeypot architectures.

The mean effect size of 0.905 strongly supports the conclusion that AI-enhanced honeypots offer superior capabilities, such as identifying zero-day threats, reducing false positives, and autonomously adapting to dynamic attack patterns. These systems are not merely incremental improvements but signify a paradigm shift from static threat monitoring to intelligent, proactive cyber deception.

However, the findings also revealed high heterogeneity ($I^2$ = 99.99%), primarily due to methodological variations and diverse deployment environments. It underscores the urgent need for standardised evaluation frameworks that ensure fair and replicable comparisons across future studies.

In terms of strategic development, while most research has focused on defensive honeypots, there is a growing interest in offensive deception systems. These actively engage attackers using reinforcement learning and adversarial modelling, further expanding the potential of AI-driven cybersecurity mechanisms.

Future research should address the following areas:

- Develop and adopt standardised benchmarking and reporting protocols.
- Prioritise real-world deployment and field validation.
- Design lightweight, scalable honeypots for constrained environments like IoT.
- Explore ethical and legal frameworks for offensive honeypot use.
- Integrate advanced AI techniques, such as XAI and GANs, to enhance system transparency and sophistication.

In conclusion, AI-based honeypots represent a transformative approach to cyber threat detection and deception. Their continued development and adoption will be instrumental in building resilient, adaptive, and intelligent security infrastructures for modern organisations. To ensure their long-term effectiveness, future research must focus not only on improving detection capabilities but also on enhancing cross-domain scalability, deployment reliability, and ethical governance. Advancing these areas will support the integration of AI-driven honeypots as standard components in proactive cybersecurity strategies.

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